CO_2 analysis in icecaps

Scientists can measure past concentrations of carbon dioxide by analyzing bubbles of air trapped in Antarctic and Greenland ice sheets. This evidence indicates that there have been several cycles of variation over the past 400,000 years. These range from lower readings in the most severe glaciations—when glaciers actually formed—to higher readings during warmer, interglacial periods. The increase since the start of the Industrial Revolution has been matched by the average global temperature. This has risen by 0.13 °F (0.07 °C) per decade since 1880 and 0.31 °F (0.17 °C) per decade since 1970.

The Intergovernmental Panel on Climate Change (IPCC) warns that unless the world's governments reduce greenhouse gas emissions dramatically, by the year 2100 average temperatures could be around 7.7 °F (4.3 °C) higher than they were before the Industrial Revolution. Such an increase would cause both a marked rise in sea levels and more extreme weather, which would result in people having to abandon some regions of the world entirely.